

DOCKHUB · DESIGN DECISIONS REGISTER

DockHub design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out: anchors and pad chosen (To confirm item 9 now the anchor rating only); value engineering restated with them
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 13 (DKH-DDR-003 accepted); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

DockHub design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Charger footprint and fixing holes, and that its power factor is 0.9 or better	The shelf holes are marked from the charger; R5 holds only with power-factor-corrected chargers	DKH-CAL-001 section 6; BOM line 5
2	Controller, MPPT and grid box sizes (no larger than 220 x 100 x 190, 150 x 70 x 180 and 220 x 92 x 200 mm)	They must fit between the floor flange and the shelf flange on the back panel	DKH-DDR-003, P9
3	Solenoid lock size (about 30 x 30 x 50 mm) and bolt throw	The tongue position on the door is set from it	DKH-DDR-003, P6
4	SwapCell v0.3 receptacle flange and fixing	The cradle's 72 x 50 x 9 mm pocket and its two M3 inserts are set from it	DKH-DDR-003, P7
5	Fans 120 x 120 x 25 mm with corner holes on a 105 mm square	The roof holes and spacer positions are set from them	DKH-DDR-003, P5
6	Aerosol unit size (about 76 x 420 mm) and its straps	Its position under the roof line, clear of the fans and beams	DKH-DDR-003, P9
7	Solar panel size, frame depth (35 mm) and that the maker allows end clamps	The rail length and clamps are set from it	DKH-DDR-003, P3
8	A printable flame-retardant filament rated UL 94 V-0	The cradle material	DKH-DDR-003, P7
9	The chosen M12 stainless wedge anchors (BOM line 17) have an approval giving a design tension resistance of at least 3.8 kN at 100 mm embedment in C25/30 concrete, 225 mm from the pad edge	R15 is met on paper with the anchors and the cast pad (BOM line 18) chosen on 2026-10-02; the anchor rating is the remaining check	DKH-CAL-001 section 11

■ Value engineering

Value-engineering target: USD 1,200. Estimated cost of the constructable design: USD 1,519 (USD 319 over the target). The target is a hypothetical control target, not a limit, for the four-bay cabinet with two bays fitted and SwapCell packs excluded. The estimate now includes the anchors and cast site pad chosen on 2026-10-02 (lines 17 and 18, USD 305); without them it is USD 1,214. All four bays fitted cost an estimated USD 1,743; each further bay adds USD 112. The vermiculite floor tray (line 19, about USD 11 a bay) is a trial option and is not in the estimate.

Main cost drivers: the site pad (line 18, USD 265: 28 bags of premixed concrete, mesh, formwork and gravel), then the parts added to make the design buildable, which took the estimate from USD 1,199 to USD 1,214 (canopy end clamps, post plates and bolts, line 12, USD 60 to USD 70; service door hinge, line 15, USD 30 to USD 35), and the anchors (line 17, USD 40). All prices are indicative.

Savings worth trying: using the host's own slab instead of the cast pad where an engineer confirms it does the same job (up to USD 265); ready-mixed concrete instead of bags if the supplier delivers a small load; recovering the USD 14 of construction parts at supplier selection.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items with a recommendation: four-bay cabinet with bays 1 and 2 fitted and a \$1,200 value-engineering target (budget_usd) for that fit; 400 W canopy kept and R6 relaxed to 10 %; four bays in one row; one charger and CAN channel per bay; steel compartment per bay with a rear plenum; doors locked during a fire alarm; MPPT to bay 1; offline-first NFC access; 70 % health gate	Amish: "i accept all your recommendations, go with them across all repos."	DKH-DDR-001, DKH-DDR-002
2026-09-25	TRL 3 engineering proposals: build to SwapCell interface v0.3; release packs at 80 % state of charge or more; chargers with a power factor of 0.9 or better; plinth flush with the body; 15 min rider patience and the demand profile of DKH-CAL-001; two-bay first prototype that cannot run the duty case	Amish, same instruction	DKH-DDR-001 items 12 and 13, DKH-DDR-002
2026-09-30	Make each design physically buildable while drawing the build plan, recording every change	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The resulting changes were accepted on 2026-10-02 (below)	DKH-DDR-003
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it."	STANDARDS section 18
2026-10-02	Design for construction accepted: the changes P1 to P12 and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	DKH-DDR-003, Table 1
2026-10-02	Bay door hinges: stainless piano hinges with the pin ends peened over (option a) for the prototype; reviewed after the first site visit	Amish: "i approve your recommendations for all 555 open decisions."	DKH-DDR-003, A2
2026-10-02	R6 restated as about 9 % of charging energy, with normal routing kept; switching the solar charger to whichever bay holds the flattest pack is to be tried as a TRL 4 controller change, not a cabinet change	Amish: "i approve your recommendations for all 555 open decisions."	DKH-DDR-002, Table 2
2026-10-02	R9: keep the pack's 45 °C charge limit and accept daytime charging pauses; R9's charging range restated as full-rate charging up to about 30 °C ambient, with the cold-soak limit near -1 °C stated; the first pilot site is chosen with summer highs near that; active cooling revisited only for a hot-climate site	Amish: "i approve your recommendations for all 555 open decisions."	DKH-DDR-002, Table 2
2026-10-02	Pack ownership: a station-owned pool of packs exchanged one for one, as the concept and R10 assume; to be confirmed in co-design	Amish: "i approve your recommendations for all 555 open decisions."	DKH-DDR-001 item 10

Date	Decision	Decided by	Record
2026-10-02	First pilot partner: a delivery-worker group in a high-income city, with New York City first and Los Deliveristas Unidos (Workers Justice Project) as the first candidate partner to approach	Amish: "i approve your recommendations for all 555 open decisions."	DKH-DDR-001 item 11
2026-10-02	Propagation test for R8: a later version aims at a UL 9540A unit-level propagation test, with a heater-triggered runaway in one bay at an outdoor test site first, before paying an accredited lab	Amish: "i approve your recommendations for all 555 open decisions."	DKH-PRC-001, Open questions
2026-10-02	Fire suppression: keep the condensed-aerosol unit for the first prototype and test vermiculite-filled bay floors as a variant in the propagation trial; choose on the result	Amish: "i approve your recommendations for all 555 open decisions."	DKH-PRC-001, Open questions
2026-10-02	Siting: the first pilot on private property with an existing single-phase circuit (the partner's own premises or a willing business frontage), not on a public sidewalk; a city sidewalk permit is sought only for a second site	Amish: "i approve your recommendations for all 555 open decisions."	DKH-PRC-001, Open questions
2026-10-02	Riders who do not want to share identity: anonymous prepaid NFC cards sold by the partner, each carrying a pack deposit, alongside named tokens	Amish: "i approve your recommendations for all 555 open decisions."	DKH-PRC-001, Open questions
2026-10-02	Product renders: solid steel doors in the build; the clear windows are shown in the renders only	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 1
2026-10-02	Product renders: scene state, header sign and markings accepted as render choices; renders captioned 'four-bay fit shown'; sign wording settled with branding	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, items 2, 3 and 5
2026-10-02	Re-render the photoreal renders, card and social preview on Amish's Mac to the constructable design, now that decisions 1 and 2 are made	Amish: "i approve your recommendations for all 555 open decisions."	DKH-DDR-003, Consequences